

Homework 3 - SSIE 553

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Question 1 [30 pts]: Simplex Method Iterations

Given LP: Min $z = -2x_1 + x_2 - x_3$

Subject to:

$$x_1 - x_2 + 3x_3 \leq 4$$

$$2x_1 + x_2 \leq 10$$

$$x_1 - x_2 - x_3 \leq 7$$

$$x_1, x_2, x_3 \geq 0$$

Step 1: Convert to Standard Form

Introduce slack variables $x_4, x_5, x_6 \geq 0$:

$$x_1 - x_2 + 3x_3 + x_4 = 4$$

$$2x_1 + x_2 + x_5 = 10$$

$$x_1 - x_2 - x_3 + x_6 = 7$$

The constraint matrix A and RHS vector b are:

$$A = \begin{bmatrix} 1 & -1 & 3 & 1 & 0 & 0 \\ 2 & 1 & 0 & 0 & 1 & 0 \\ 1 & -1 & -1 & 0 & 0 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 4 \\ 10 \\ 7 \end{bmatrix}$$

Cost vector: $c = [-2, 1, -1, 0, 0, 0]^T$

Iteration k = 0 (Initial BFS)

Initial basic variables are the slack variables. Non-basic variables are set to zero.

Basic and Non-basic Variables:

$$x_{B_0} = [x_4, x_5, x_6]^T = [4, 10, 7]^T$$

$$x_{N_0} = [x_1, x_2, x_3]^T$$

Basis Matrix B_0 and Non-basic Matrix N_0 :

$$B_0 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad N_0 = \begin{bmatrix} 1 & -1 & 3 \\ 2 & 1 & 0 \\ 1 & -1 & -1 \end{bmatrix}$$

Simplex Multipliers (λ_0):

$$c_B = [0, 0, 0]^T$$

$$\lambda_0 = c_B^T * B_0^{-1} = [0, 0, 0] * I = [0, 0, 0]$$

Reduced Costs (d_0):

$$c_N = [-2, 1, -1]^T$$

$$d_0 = c_N - N_0^T * \lambda_0 = [-2, 1, -1] - [0, 0, 0] = [-2, 1, -1]$$

Since $d_1 = -2 < 0$ (most negative), x_1 enters the basis (minimization: enter most negative reduced cost).

Ratio Test (Minimum Ratio for x_1 entering):

$$B_0^{-1} * a_1 = I * [1, 2, 1]^T = [1, 2, 1]^T$$

$$\text{Ratios: } 4/1 = 4, \quad 10/2 = 5, \quad 7/1 = 7$$

$$\text{Min ratio} = 4 \text{ (row 1)} \Rightarrow x_4 \text{ leaves the basis}$$

Iteration k = 1 (After First Pivot)

Pivot: x_1 enters, x_4 leaves. Pivot element = 1 at position (row 1, col 1).

Computing B_1^{-1} :

$$B_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad (\text{columns of } A \text{ for } x_1, x_5, x_6)$$

Row-reduce $[B_1 \mid I]$:

$$R_2 = R_2 - 2 \cdot R_1: [0, 1, 0 \mid -2, 1, 0]$$

$$R_3 = R_3 - 1 \cdot R_1: [0, 0, 1 \mid -1, 0, 1]$$

$$B_1^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Basic and Non-basic Variables:

$$\begin{aligned} x_{B_1} &= B_1^{-1} \cdot b = [1, 0, 0; -2, 1, 0; -1, 0, 1] \cdot [4; 10; 7] \\ &= [4, -8+10, -4+7] = [4, 2, 3] \end{aligned}$$

$$x_{B_1} = [x_1, x_5, x_6]^T = [4, 2, 3]^T$$

$$x_{N_1} = [x_4, x_2, x_3]^T$$

Basis Matrix B_1 and Non-basic Matrix N_1 :

$$B_1 = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \quad N_1 = \begin{bmatrix} 1 & -1 & 3 \\ 0 & 1 & 0 \\ 0 & -1 & -1 \end{bmatrix}$$

Simplex Multipliers (λ_1):

$$c_B = [-2, 0, 0]^T \quad (\text{costs for } x_1, x_5, x_6)$$

$$\begin{aligned} \lambda_1 &= c_B^T \cdot B_1^{-1} \\ &= [-2, 0, 0] \cdot [1, 0, 0; -2, 1, 0; -1, 0, 1] \\ &= [-2, 0, 0] \end{aligned}$$

Reduced Costs (d_1):

$$c_N = [0, 1, -1]^T \quad (\text{costs for } x_4, x_2, x_3)$$

$$\begin{aligned} \lambda_1 \cdot N_1 &= [-2, 0, 0] \cdot [1, -1, 3; 0, 1, 0; 0, -1, -1] \\ &= [-2, 2, -6] \end{aligned}$$

$$d_1 = c_N - (\lambda_1 \cdot N_1)^T = [0, 1, -1] - [-2, 2, -6] = [2, -1, 5]$$

Since $d_2 = -1 < 0$ (for x_2), x_2 enters the basis.

Ratio Test (x_2 entering):

$$\begin{aligned} B_1^{-1} \cdot a_2 &= [1, 0, 0; -2, 1, 0; -1, 0, 1] \cdot [-1; 1; -1] \\ &= [-1, 3, 0]^T \end{aligned}$$

Only positive entry: row 2 (value 3)

Ratio: $2/3$

Min ratio = $2/3$ (row 2) $\Rightarrow x_5$ leaves the basis

Iteration k = 2 (Optimal Solution)

Pivot: x2 enters, x5 leaves.

Computing B_2^{-1} :

$$B_2 = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix} \quad (\text{columns of A for } x_1, x_2, x_6)$$

Row-reduce $[B_2 \mid I]$:

$$\begin{aligned} R_2 &= R_2 - 2R_1: & [0, \quad 3, \quad 0 \mid -2, \quad 1, \quad 0] \\ R_3 &= R_3 - R_1: & [0, \quad 0, \quad 1 \mid -1, \quad 0, \quad 1] \\ R_2 &= R_2/3: & [0, \quad 1, \quad 0 \mid -2/3, \quad 1/3, \quad 0] \\ R_1 &= R_1 + R_2: & [1, \quad 0, \quad 0 \mid 1/3, \quad 1/3, \quad 0] \end{aligned}$$

$$B_2^{-1} = \begin{bmatrix} 1/3 & 1/3 & 0 \\ -2/3 & 1/3 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

Basic and Non-basic Variables:

$$\begin{aligned} xB_2 &= B_2^{-1} \cdot b \\ &= [1/3, 1/3, 0; -2/3, 1/3, 0; -1, 0, 1] \cdot [4; 10; 7] \\ &= [4/3 + 10/3, -8/3 + 10/3, -4 + 7] \\ &= [14/3, 2/3, 3] \\ xB_2 &= [x_1, x_2, x_6]^T = [14/3, 2/3, 3]^T \\ xN_2 &= [x_4, x_5, x_3]^T \end{aligned}$$

Basis Matrix B_2 and Non-basic Matrix N_2 :

$$B_2 = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 1 & 0 \\ 1 & -1 & 1 \end{bmatrix}, \quad N_2 = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{bmatrix}$$

Simplex Multipliers (λ_2):

$$\begin{aligned} cB &= [-2, 1, 0]^T \quad (\text{costs for } x_1, x_2, x_6) \\ \lambda_2 &= cB^T \cdot B_2^{-1} \\ &= [-2, 1, 0] \cdot [1/3, 1/3, 0; -2/3, 1/3, 0; -1, 0, 1] \\ &= [-2/3 - 2/3, -2/3 + 1/3, 0] \\ &= [-4/3, -1/3, 0] \end{aligned}$$

Reduced Costs (d_2):

$$\begin{aligned} cN &= [0, 0, -1]^T \quad (\text{costs for } x_4, x_5, x_3) \\ \lambda_2 \cdot N_2 &= [-4/3, -1/3, 0] \cdot [1, 0, 3; 0, 1, 0; 0, 0, -1] \\ &= [-4/3, -1/3, -4] \\ d_2 &= cN - (\lambda_2 \cdot N_2)^T = [0, 0, -1] - [-4/3, -1/3, -4] \\ &= [4/3, 1/3, 3] \end{aligned}$$

Optimality Check:

All reduced costs $d_2 = [4/3, 1/3, 3] \geq 0$. For minimization, this means no improvement is possible. The current BFS is OPTIMAL.

Optimal Solution:

$$x_1^* = 14/3, \quad x_2^* = 2/3, \quad x_3^* = 0$$

$$z^* = -2(14/3) + 1(2/3) - 1(0) = -28/3 + 2/3 = -26/3$$

Question 2 [20+5 pts]: Simplex Method & Alternative Optima

Given LP: $\text{Max } z = x_1 + x_2$

Subject to:

$$x_1 + x_2 + x_3 \leq 1$$

$$x_1 + 2x_3 \leq 1/2$$

$$x_1, x_2, x_3 \geq 0$$

Part (i): Solve using the Simplex Method

Standard form with slack variables $x_4, x_5 \geq 0$:

$$x_1 + x_2 + x_3 + x_4 = 1$$

$$x_1 + 2x_3 + x_5 = 1/2$$

Initial Simplex Tableau:

BV	x1	x2	x3	x4	x5	RHS
x4	1	1	1	1	0	1
x5	1	0	2	0	1	1/2
z	-1	-1	0	0	0	0

For maximization, enter variable with most negative z-row coefficient. Both x_1 and x_2 have coefficient -1. Choose x_1 (leftmost).

Ratio Test (x_1 entering):

$$\text{Row 1: } 1/1 = 1$$

$$\text{Row 2: } (1/2)/1 = 1/2 \quad \leftarrow \text{minimum}$$

x_5 leaves. Pivot element = 1 at (row 2, col 1).

Pivot Operations:

$$\text{New R2} = \text{R2 (pivot} = 1, \text{ unchanged)}$$

$$\text{New R1} = \text{R1} - \text{R2: } [0, 1, -1, 1, -1 \mid 1/2]$$

$$\text{New z} = \text{z} + \text{R2: } [0, -1, 2, 0, 1 \mid 1/2]$$

Tableau After Iteration 1:

BV	x1	x2	x3	x4	x5	RHS
x4	0	1	-1	1	-1	1/2
x1	1	0	2	0	1	1/2
z	0	-1	2	0	1	1/2

x_2 has coefficient -1 (negative). x_2 enters.

Ratio Test (x_2 entering):

$$\text{Row 1: coeff} = 1 > 0, \text{ ratio} = (1/2)/1 = 1/2$$

$$\text{Row 2: coeff} = 0, \text{ skip}$$

x_4 leaves. Pivot element = 1 at (row 1, col 2).

Pivot Operations:

New R1 = R1 (pivot = 1, unchanged)
New R2 = R2 (coeff of x2 = 0, unchanged)
New z = z + R1: [0, 0, 1, 1, 0 | 1]

Tableau After Iteration 2 (Optimal):

BV	x1	x2	x3	x4	x5	RHS
x2	0	1	-1	1	-1	1/2
x1	1	0	2	0	1	1/2
z	0	0	1	1	0	1

All z-row coefficients ≥ 0 . OPTIMAL for maximization.

Optimal Solution: $x1^* = 1/2$, $x2^* = 1/2$, $x3^* = 0$, $z^* = 1$

Part (ii): Alternative Optimal Solution?

Rule: An alternative optimal solution exists if a NON-BASIC variable has a reduced cost of exactly zero in the optimal tableau.

Examining the z-row of the optimal tableau:

Non-basic variables: x_3 (coeff = 1), x_4 (coeff = 1), x_5 (coeff = 0)

x_5 has a reduced cost of exactly 0. YES, an alternative optimal solution exists.

To find the alternative extreme point, bring x_5 into the basis:

Column for x_5 in current tableau: $[-1, 1]^T$

Ratio test:

Row 1: coeff = -1 (negative, skip)

Row 2: coeff = 1, ratio = $(1/2)/1 = 1/2$

x_1 leaves. Pivot on (row 2, col 5), pivot = 1.

Pivot Operations:

New R2 = R2: $[1, 0, 2, 0, 1 \mid 1/2]$

New R1 = R1 + R2: $[1, 1, 1, 1, 0 \mid 1]$

z-row unchanged (coeff of x_5 = 0)

New Optimal Tableau:

BV	x_1	x_2	x_3	x_4	x_5	RHS
----	-----	-----	-----	-----	-----	-----
x_2	1	1	1	1	0	1
x_5	1	0	2	0	1	$1/2$
----	-----	-----	-----	-----	-----	-----
z	0	0	1	1	0	1

Alternative Optimal Extreme Point:

$x_1^* = 0$, $x_2^* = 1$, $x_3^* = 0$, $z^* = 1$

Both solutions give the same optimal value $z^* = 1$, confirming alternative optima.

Question 3 [5+20 pts]: Unbounded LP

Given LP: $\text{Max } z = 2x_2$

Subject to:

$$x_1 - x_2 \leq 4$$

$$-x_1 + x_2 \leq 1$$

$$x_1, x_2 \geq 0$$

Part (i): State the Unboundedness Rule

Unboundedness Rule:

During the simplex method for a maximization problem, if a non-basic variable x_j is selected to enter the basis (because its reduced cost is positive, i.e., $c_j - z_j > 0$), and ALL entries in the corresponding pivot column are less than or equal to zero ($y_{ij} \leq 0$ for all rows i), then there is no finite upper bound on how much x_j can increase. Consequently, the objective function z can be increased without bound, and the LP is UNBOUNDED.

Part (ii): Show the LP is Unbounded Using the Rule

Standard form with slack variables $x_3, x_4 \geq 0$:

$$x_1 - x_2 + x_3 = 4$$

$$-x_1 + x_2 + x_4 = 1$$

Initial Simplex Tableau:

BV	x_1	x_2	x_3	x_4	RHS
----	-----	-----	-----	-----	-----
x_3	1	-1	1	0	4
x_4	-1	1	0	1	1
----	-----	-----	-----	-----	-----
z	0	-2	0	0	0

For Max, x_2 has z -row coefficient -2 (most negative). x_2 enters.

Ratio Test (x_2 entering):

Pivot column for x_2 : $[-1, 1]^T$

Row 1: coeff = -1 (negative, skip)

Row 2: coeff = 1, ratio = $1/1 = 1$

x_4 leaves. Pivot element = 1 at (row 2, col 2).

Pivot Operations:

New $R_2 = R_2$: $[-1, 1, 0, 1 \mid 1]$

New $R_1 = R_1 + R_2$: $[0, 0, 1, 1 \mid 5]$

New $z = z + 2 \cdot R_2$: $[-2, 0, 0, 2 \mid 2]$

Tableau After Iteration 1:

BV	x_1	x_2	x_3	x_4	RHS
----	-----	-----	-----	-----	-----
x_3	0	0	1	1	5
x_2	-1	1	0	1	1
----	-----	-----	-----	-----	-----
z	-2	0	0	2	2

For Max, x_1 has z-row coefficient -2 (negative), so x_1 should enter the basis.

Check the Pivot Column for x_1 :

Column for x_1 : $[0, -1]^T$

Applying the Unboundedness Rule:

ALL entries in the x_1 column are ≤ 0 :

Row 1: $0 \leq 0$ (zero)

Row 2: $-1 \leq 0$ (negative)

Since we want to bring x_1 into the basis (it would improve the objective), but ALL entries in the pivot column are non-positive, there is no valid pivot. We can increase x_1 without limit, and since x_1 has a negative coefficient in the z-row (-2), the objective z will increase without bound.

Conclusion: This LP is UNBOUNDED. The objective function $z = 2x_2$ can be made arbitrarily large.

Question 4 [10+5+5 pts]: Degeneracy in LP

Given LP: $\text{Max } z = 5x_1 + 3x_2$

Subject to:

$$4x_1 + 2x_2 \leq 12$$

$$4x_1 + x_2 \leq 10$$

$$x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

Part (i): Solve Using Simplex Until Degenerate BFS

Standard form with slack variables $x_3, x_4, x_5 \geq 0$:

$$4x_1 + 2x_2 + x_3 = 12$$

$$4x_1 + x_2 + x_4 = 10$$

$$x_1 + x_2 + x_5 = 4$$

Initial Tableau (Iteration 0):

BV	x1	x2	x3	x4	x5	RHS
x3	4	2	1	0	0	12
x4	4	1	0	1	0	10
x5	1	1	0	0	1	4
z	-5	-3	0	0	0	0

Most negative z-row coefficient: x_1 (-5). x_1 enters.

Ratio Test:

$$\text{Row 1: } 12/4 = 3$$

$$\text{Row 2: } 10/4 = 2.5 \leftarrow \text{minimum}$$

$$\text{Row 3: } 4/1 = 4$$

x_4 leaves. Pivot at (row 2, col 1), pivot = 4.

Pivot Operations:

$$\text{New R2} = \text{R2}/4: [1, 1/4, 0, 1/4, 0 \mid 5/2]$$

$$\text{New R1} = \text{R1} - 4 \cdot \text{R2}: [0, 1, 1, -1, 0 \mid 2]$$

$$\text{New R3} = \text{R3} - 1 \cdot \text{R2}: [0, 3/4, 0, -1/4, 1 \mid 3/2]$$

$$\text{New z} = \text{z} + 5 \cdot \text{R2}: [0, -7/4, 0, 5/4, 0 \mid 25/2]$$

Tableau After Iteration 1:

BV	x1	x2	x3	x4	x5	RHS
x3	0	1	1	-1	0	2
x1	1	1/4	0	1/4	0	5/2
x5	0	3/4	0	-1/4	1	3/2
z	0	-7/4	0	5/4	0	25/2

x_2 has z-row coefficient -7/4 (negative). x_2 enters.

Ratio Test:

Row 1: $2/1 = 2$

Row 2: $(5/2)/(1/4) = 10$

Row 3: $(3/2)/(3/4) = 2 \quad \leftarrow \text{TIE!}$

Tie between Row 1 and Row 3 (both ratio = 2)

=> This TIE signals DEGENERACY!

Choose Row 1: x3 leaves.

Pivot Operations (x2 enters, x3 leaves):

Pivot element = 1 at (row 1, col 2)

New R1 = R1: $[0, 1, 1, -1, 0 \mid 2]$

New R2 = R2 - (1/4)*R1: $[1, 0, -1/4, 1/2, 0 \mid 2]$

New R3 = R3 - (3/4)*R1: $[0, 0, -3/4, 1/2, 1 \mid 0]$

New z = z + (7/4)*R1: $[0, 0, 7/4, -3/4, 0 \mid 16]$

Tableau After Iteration 2 (Degenerate BFS):

BV	x1	x2	x3	x4	x5	RHS
x2	0	1	1	-1	0	2
x1	1	0	-1/4	1/2	0	2
x5	0	0	-3/4	1/2	1	0
z	0	0	7/4	-3/4	0	16

Degenerate BFS Confirmed:

The basic variable $x_5 = 0$ in the basis. A BFS is degenerate when one or more basic variables equal zero. This occurred because of the tie in the ratio test at Iteration 1.

Current BFS: $x_1 = 2, x_2 = 2, x_3 = 0, x_4 = 0, x_5 = 0, z = 16$

Note: x_4 has z-row coefficient -3/4 (negative), so this is not yet optimal. However, the question only asks us to solve until we confirm degeneracy, which we have done.

Part (ii): Graph the Feasible Region

At the degenerate BFS $(x_1, x_2) = (2, 2)$, we check which constraints are binding:

Constraint 1: $4(2) + 2(2) = 12 = 12 \Rightarrow$ BINDING

Constraint 2: $4(2) + 1(2) = 10 = 10 \Rightarrow$ BINDING

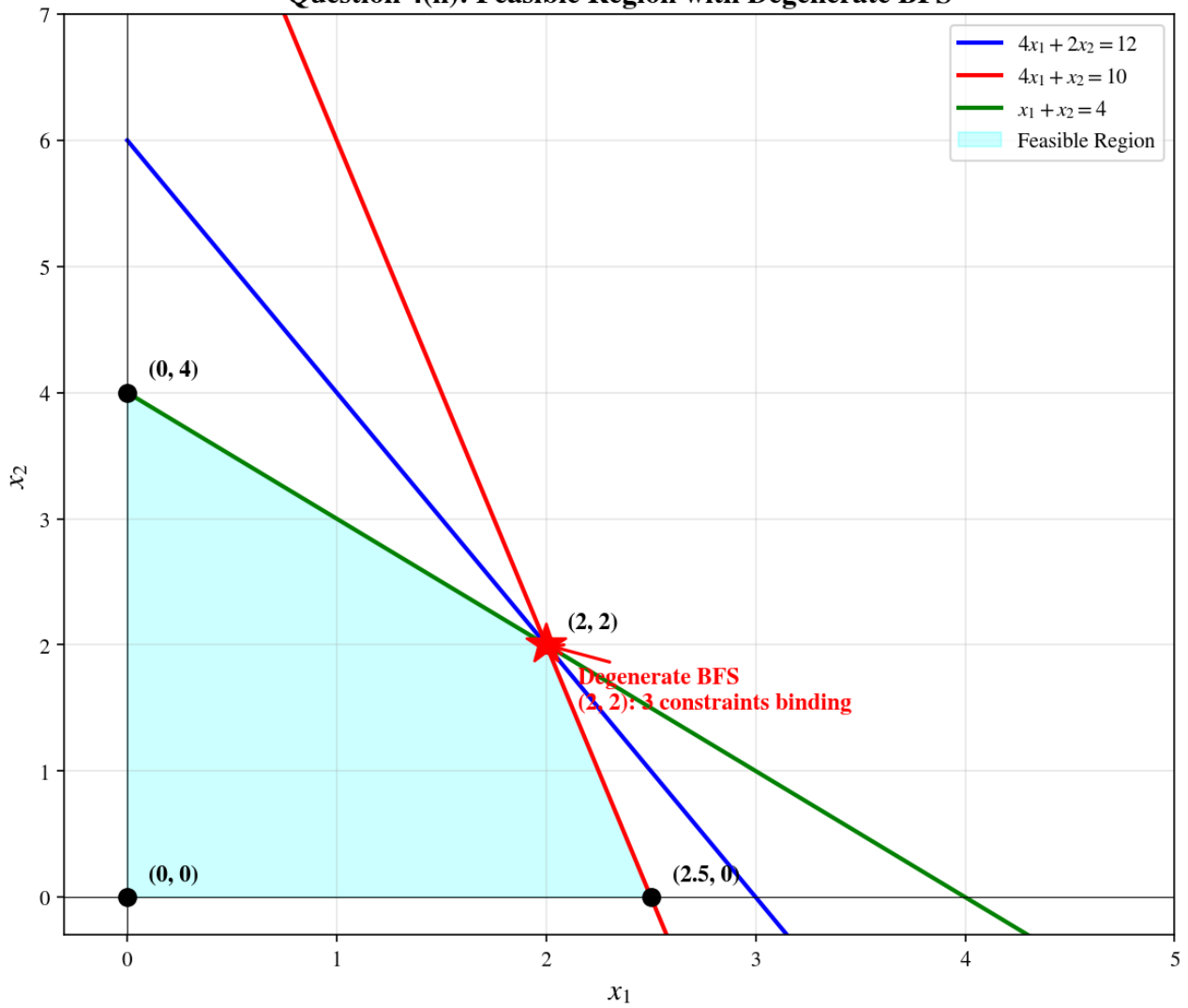
Constraint 3: $1(2) + 1(2) = 4 = 4 \Rightarrow$ BINDING

All THREE original constraints are binding at the point (2, 2).

This is the geometric signature of degeneracy: more constraints meet at this vertex than needed to define it (only 2 are needed in 2D).

In standard form, the binding constraints correspond to the non-negativity constraints $x_3 \geq 0, x_4 \geq 0$, and $x_5 \geq 0$ all being active (all three equal zero). See the graph below:

Question 4(ii): Feasible Region with Degenerate BFS



Part (iii): Are Binding Constraints Greater Than n ?

In standard form $Ax = b$, $x \geq 0$, the key quantities are:

$n = 5$ (total variables: x_1, x_2, x_3, x_4, x_5)

$m = 3$ (number of equality constraints)

Count of binding constraints at the degenerate BFS ($x_1=2, x_2=2, x_3=0, x_4=0, x_5=0$):

- 3 equality constraints ($Ax = b$): always binding
- 3 non-negativity constraints binding: $x_3=0, x_4=0, x_5=0$

Total binding constraints = $3 + 3 = 6$

Is $6 > n = 5$? YES!

The number of binding constraints (6) exceeds the number of variables $n = 5$. For a non-degenerate BFS, exactly $n = 5$ constraints should be binding ($m = 3$ equality constraints + exactly $n - m = 2$ active non-negativity constraints). Here we have 3 active non-negativity constraints instead of 2, which is the algebraic definition of degeneracy.